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Abstract
In this paper, the possible evolution of a pinched plasma column is presented from the results of temporally resolved measurements using a magnetic probe, interferometry and neutron diagnostics performed on the plasma focus PF-1000 device with deuterium as the filling gas. Together with the discharge axial current of about 1.5 MA a toroidal current component of the order of 100 kA was estimated in the toroidal, helical and plasmoidal structures formed within the dense plasma column. The mass inside these structures increases due to injection of the plasma from the neighborhood regions with a higher pinching pressure. This injected plasma increases the intensity of the internal magnetic field, probably through turbulent motion and the magnetic dynamo effect. The neutrons from the D–D fusion reaction, produced during the formation and decay of plasmoidal structures and constrictions, are accompanied by changes in the axial component of the magnetic field. Then, the transformation and decay of internal closed currents can contribute to the acceleration of high-energy electrons and ions.

1. Introduction

A plasma focus device, as a type of Z-pinch discharge, is a simple source of dense and hot plasma convenient for the study of astrophysical and fusion-related phenomena [1–5]. The long lifetime of the pinch phase and the production of high-energy electron and ion beams with energy of hundreds of keV present special features that are common for the pinched plasma observed on a wide scale of densities, times and dimensions. Non-thermal mechanisms of high-energy particle acceleration are a subject of discussion [6, 7], mainly the influence of the evolution of instabilities, microturbulences, filamentary structure, anomalous resistance, distribution of internal currents and transformations of magnetic fields.

Recently, on the plasma focus PF-1000 device the plasma has been investigated with laser interferometry, x-ray and neutron diagnostics [8]. Interferometry showed the existence of several toroidal, helical and plasmoidal structures inside the plasma column. These structures are usually observed during all phases of pinch evolution: the imploding current sheath, the stagnation and the phase of instabilities. Intense high-energy electrons and ions are produced in correlation with the formation and disintegration of plasmoids and constrictions. The signals of magnetic probes show the presence of azimuthal (B_φ) and axial (B_z) components of the magnetic field during the pinch evolution [9, 10]. The current sheath imploding the dense plasma layer contains the majority of the total discharge current and it passes outside the dense plasma layer. It is composed of the B_φ and also the B_z component with the amplitude roughly 2–8 times lower.

In [11] the results from interferometry and neutron diagnostics were correlated with the magnetic probe signals. The results showed a close connection of evolution of the dense toroidal, helical and spherical structures with changes in the axial component of the magnetic field and neutron production. This axial magnetic field reached a value of up to 2–8 T, being 10–30% of the azimuthal magnetic field existing on the surface of the pinched plasma column and which can considerably influence the dynamics of the pinch and internal energy transformations.

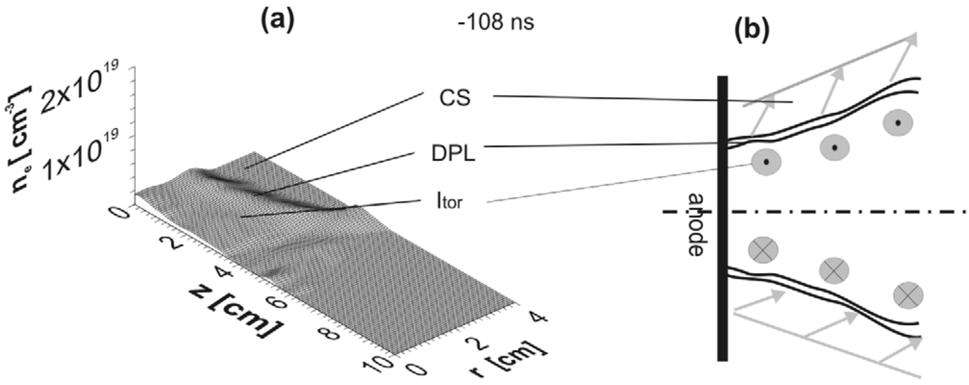

Figure 1. (*a*) Shot 9354, densitogram registered at −108 ns, (*b*) scheme of the imploding dense plasma layer (DPL) and possible current distribution in the current sheath (CS) and in the layer in front of the DPL (I_{tor}); r is the radial distance from the axis and $z = 0$ is the position of the anode face.

In this paper, we summarize the experimental results from PF-1000 and formulate the evolution of the structures in the dense pinch on the basis of the transformations of internal closed currents associated with registered signals of magnetic probes at the phases of implosion, stagnation and evolution of instabilities in correlation with the transport of the plasma, acceleration of fast particles and neutron production.

2. Description of the device and diagnostics

The measurements described here were carried out within the PF-1000 facility equipped with Mather-type coaxial electrodes of length 480 mm and an anode of diameter 230 mm. The cathode of diameter 400 mm was composed of twelve 82 mm-o.d. stainless-steel rods distributed symmetrically around the anode circumference. The capacitor bank was charged to a voltage of 21–24 kV, which corresponded to discharge energies of 290–380 kJ. The discharge current during the pinch phase amounted to 1.2–1.8 MA. The initial pressure of the pure deuterium filling was 200–600 Pa and the total neutron yield was at the level of 10^{11} per shot.

The signals of the voltage, the current derivative in time and the current were measured with the current collector near the insulator at the bottom of the anode. Soft x-ray (SXR) pulses in the range of photon energy 0.6–15 keV were recorded with a silicon PIN detector shielded by means of a Be foil of 10 μm thickness. To determine the time of generation of the hard x-rays (HXRs) and neutrons, energy distributions of registered neutrons and the mean energy of deuterons producing neutrons by the time-of-flight method [12], we used five scintillation detectors coupled with fast photomultipliers situated side-on (distance 7 m) and upstream (distances 7, 21, 50 and 84 m). The interferometric measurements were made with a Nd : YLF laser operated at the second harmonics (527 nm). The laser pulse (lasting less than 1 ns) was split by a set of mirrors into 16 separated beams, which passed through a Mach–Zehnder interferometer. These beams investigated the plasma region with a delay ranging from 0 to 210 ns [13]. The total neutron yield was calculated from the data recorded with four silver-activation counters. The temporal position of the current derivative dip was assigned as $t = 0$.

The azimuthal and axial components of the magnetic field were recorded with absolute calibrated probes placed near the anode face at radii of 4, 1.3 and 0 cm. Each probe measured the time derivative of the azimuthal (dB_φ) and axial (dB_z) component of the magnetic field. Each B_z probe coil was calibrated for its sensitivity to B_φ and this influence was subtracted during signal processing to obtain the pure dB_zdt signal [10].

3. Description of the different phases of evolution of the dense pinched column

3.1. Implosion of the dense plasma layer

As was mentioned in [9, 10, 11], the current sheath contains 80 ± 20% of the total discharge current. The width of the current sheath was estimated to be 1.6–2.6 cm. The current passed through the region outside the dense plasma layer with a density below 1×10^{24} m^{-3}. The current sheath was composed of both B_φ and B_z components, in which the amplitude of B_z of 1–2 T was roughly 2–8 times lower than B_φ. In front of the dense plasma layer, a layer with a weak B_z prepulse (about 0.1 T) was registered with opposite sign in comparison with the orientation observed in the main current sheath. The orientation of B_z lines in the precursor was directed away from the anode, while the orientation of B_z lines in the current sheath was directed in the opposite way, to the anode.

B_z as a component of the poloidal magnetic field has closed lines: consequently, B_z at a larger radial distance must have the opposite sign as well. However, the opposite B_z in the main current sheath was not registered by the probes. It could be located above the anode radial distance of 4 cm from the axis, as was discussed in [14]. The opposite sign of B_z of the precursor can be contained in the main current sheath.

The orientations of B_z and B_φ and their current in the current sheath have a helical-like left side orientation. The total toroidal component of the current can be estimated as 60–250 kA.

The positions of both current layers and the dense plasma during implosion are imaged in figure 1(*b*) assuming the existence of a similar current distribution at higher z-distances, as was measured by probes near the anode. In figure 1(*a*), we

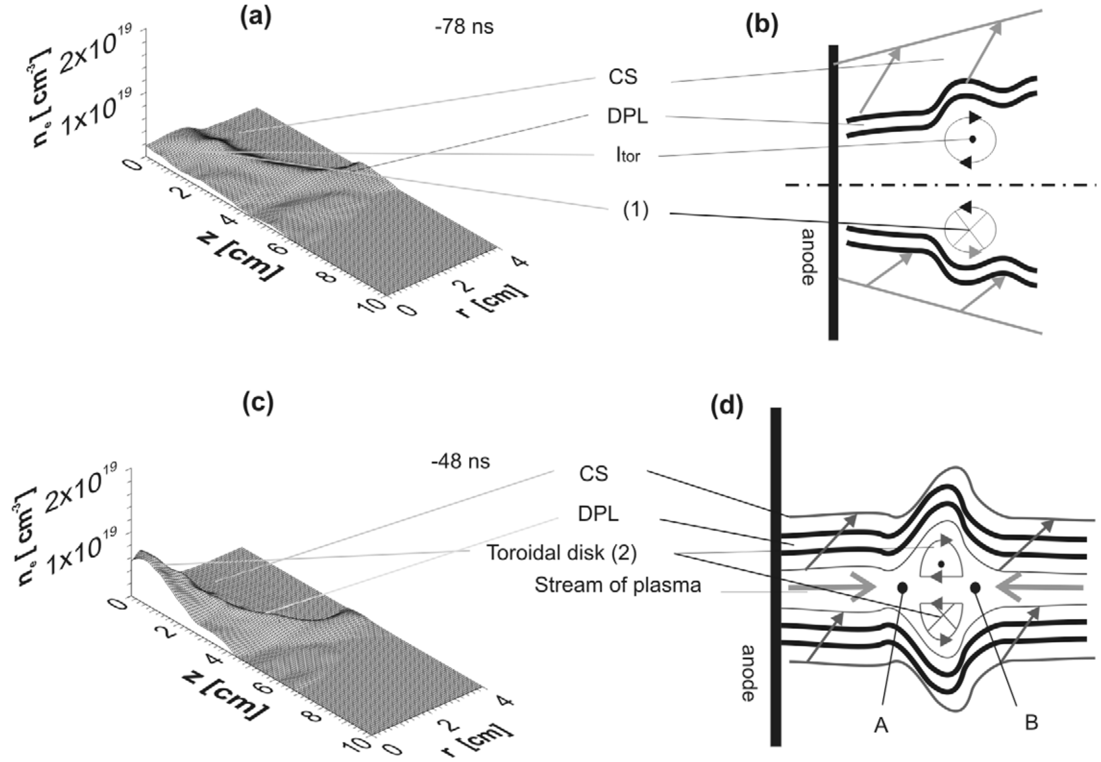

Figure 2. (*a*) Shot 9354: formation of the toroidal structure (1), (*b*) scheme of toroidal current with stabilized poloidal component (I_{tor}), (*c*) shot 9354: formation of the toroidal disk (2), (*d*) the scheme of the motion of the streams of plasma increasing both the plasma density and the closed internal toroidal and poloidal currents in the axis region of the toroidal structure.

can see the picture of the plasma density calculated by the Abel inversion supposing the plasma to have a radial symmetry.

3.2. Formation of the toroidal, helical and disk structures

In most shots, the uniformity of the imploded dense plasma column is disrupted, especially at a higher initial pressure of deuterium in the chamber. Usually, the toroidal and helical structures emerge in the interferometry images. They can considerably deform the surface of the dense layer and form lobules in different axial positions. The azimuthal length of the lobules can range from the shortest of radius dimension (taking the periphery into account) to long values (a few circumferences) in the helical form retain the intended meaning]. The most developed toroidal lobule forms at the heel of the umbrella shape of the dense plasma sheath. The external end of this lobule is usually directed to the anode.

The birth of the toroidal or helical structures correlates with the start of the B_z signal registered at the anode center (we can see an example in figures 2(*a*) and (*b*)). It corresponds to the prepulse described in section 3.1. B_z in the axis of the toroidal structures increases to 1–2 T [11] and the toroidal current (for diameter of about 2 cm) reaches a value of 100 kA. The long lifetime of the toroidal structures of the order of hundred ns shows the possibility of the existence of a stabilized component of the internal toroidal current. Hence, the current in the toroidal structure should have a helical shape with B_z in its axial region. The implosion of the region of the toroidal structure is slower than the implosion of the neighbor regions. In the meantime, the penetration of the current into the dense plasma also starts and the toroidal currents defend this movement. The pinching magnetic pressure produced in the axis locality of the neighboring regions is higher than the pressure at the center of the toroid. For example, in figure 2(*c*), the diameter of the dense column with a toroidal structure of 3.2 cm is twice as high as the diameter of the column near the anode; then the pressure in the toroidal axial region can be a few times lower. From the location with a higher pinching pressure, the plasma streams are injected into the toroidal axis, where they are captured and they increase the axial plasma density. Then, the toroidal structure is transformed into a toroidal disk with maximum density in its axial region (figures 2(*c*) and (*d*)).

An example can be seen in figure 2. The orientation of the toroidal current in the toroidal disk is determined by the sign of the registered B_z, the same as in the prepulse. If we assume the orientation of the stabilized poloidal current in the toroidal external layer to be the same as in the neighboring current sheaths, then the direction in the axial region is directed to the anode opposite to the orientation of the discharge current. Points A and B represent the zero of the magnetic field and the locality of the splitting of the magnetic streamlines. The plasma streams from the neighboring regions of the toroidal structures are apparently confined by the magnetic fields inside the toroidal locality, generate turbulent structures and increase the internal energy through reconnection and the dynamo effect, which increase the internal currents and specifically its poloidal component.

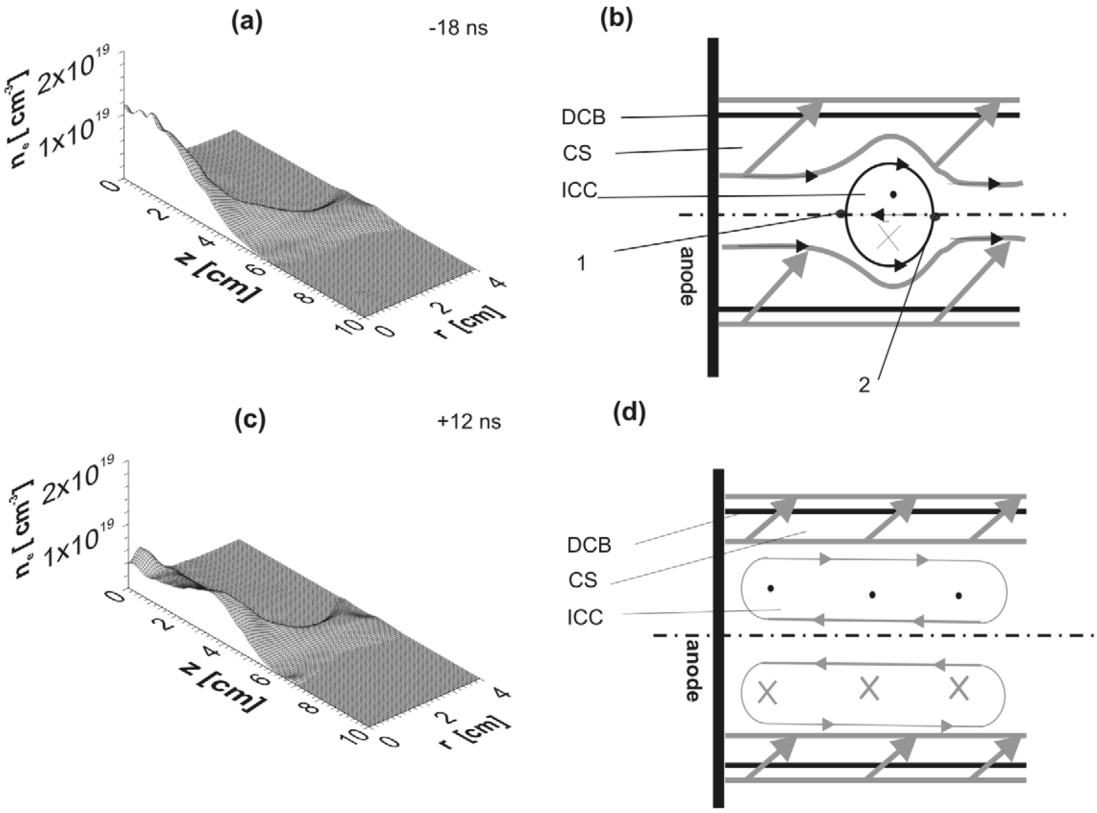

Figure 3. (*a*) Shot 9354: densitogram of the pinched column with plasmoid, (*b*) possible distribution of the current in the plasma column and in the plasmoid, (*c*) shot 9354: densitogram of the pinched column after the first neutron pulse when the plasma is transported to the column boundary, (*d*) possible distribution of the current in the plasma column after the first neutron pulse. The dense column boundary indicated here is DCB and the internal closed current ICC.

3.3. The first neutron pulse

At later evolution of the implosion, the plasma column reaches the minimal diameter first near the anode. The prominence of the toroidal-like structures on the column surface is reduced and the plasma from the thin dense layer is transported radially to the column axis. This transport is caused probably through the penetration of the current sheath into the column (registered with B_φ probes at 1.3 cm distance). In this evolution, B_φ and its magnetic pressure increase in the axis region of the pinch and support the continuation of the injection of the plasma into the dense toroidal disk and the formation of the plasmoid. This phase correlates with the emission of SXRs and with the onset of the first neutron pulse and HXRs. Likewise, the value of the B_z signal registered in the axial probe position increases and reaches the maximum above 1–2 T at the time of the maximum of the plasma density in the plasmoid and HXR intensity. The radial and axial profiles of the electron density in the plasmoid have a Gaussian shape [8]. This distribution is probably caused by the current distribution causing the plasmoid to assume a spherical shape. The minimum of the current derivative at about 20 ns after the maximum of SXRs corresponds to the time of penetration of the discharge current into the dense column and the increase in inductance of the discharge circuit. The 10–20 ns delay of the maximum of the first neutron pulse after HXR corresponds to the time between fast deuteron acceleration and neutron production at a distance of 3–10 cm. Since the dense plasmoid should be the dominant source of neutrons, the locality of deuteron acceleration could be the region around the magnetic zero point A in figure 2. This suggestion agrees with the idea of reconnection of magnetic lines as the mechanism contributing to the acceleration of deuterons.

After reaching the maximum, the B_z signal decreases together with the start and continuation of neutron production and the expansion of the column diameter. The plasmoid increases its dimension, its density decreases and it moves from the anode to the dominant toroidal lobule in which its own plasmoidal disk and plasmoid form. Due to plasmoidal higher density, it is an important source of fusion neutrons (figure 4).

At the final phase of stagnation, the plasma moves back from the axial region to the column boundaries, where the density gradient increases, and in the central region of the plasma column the plasma density of $10^{24}\,\mathrm{m}^{-3}$ is constant along the z-axis. This picture can be interpreted with redistribution of the current with the toroidal component to the column boundary with B_z displayed uniformly along the internal layers of this column (figures 3(*c*) and (*d*)).

3.4. Formation of instabilities and dominant neutron production

The evolution of the $m = 0$ instability starts together with the axial variation of densities in the stagnated column, which are initially uniform. The localities with lowering density implode and form necks or constrictions. In the toroidal

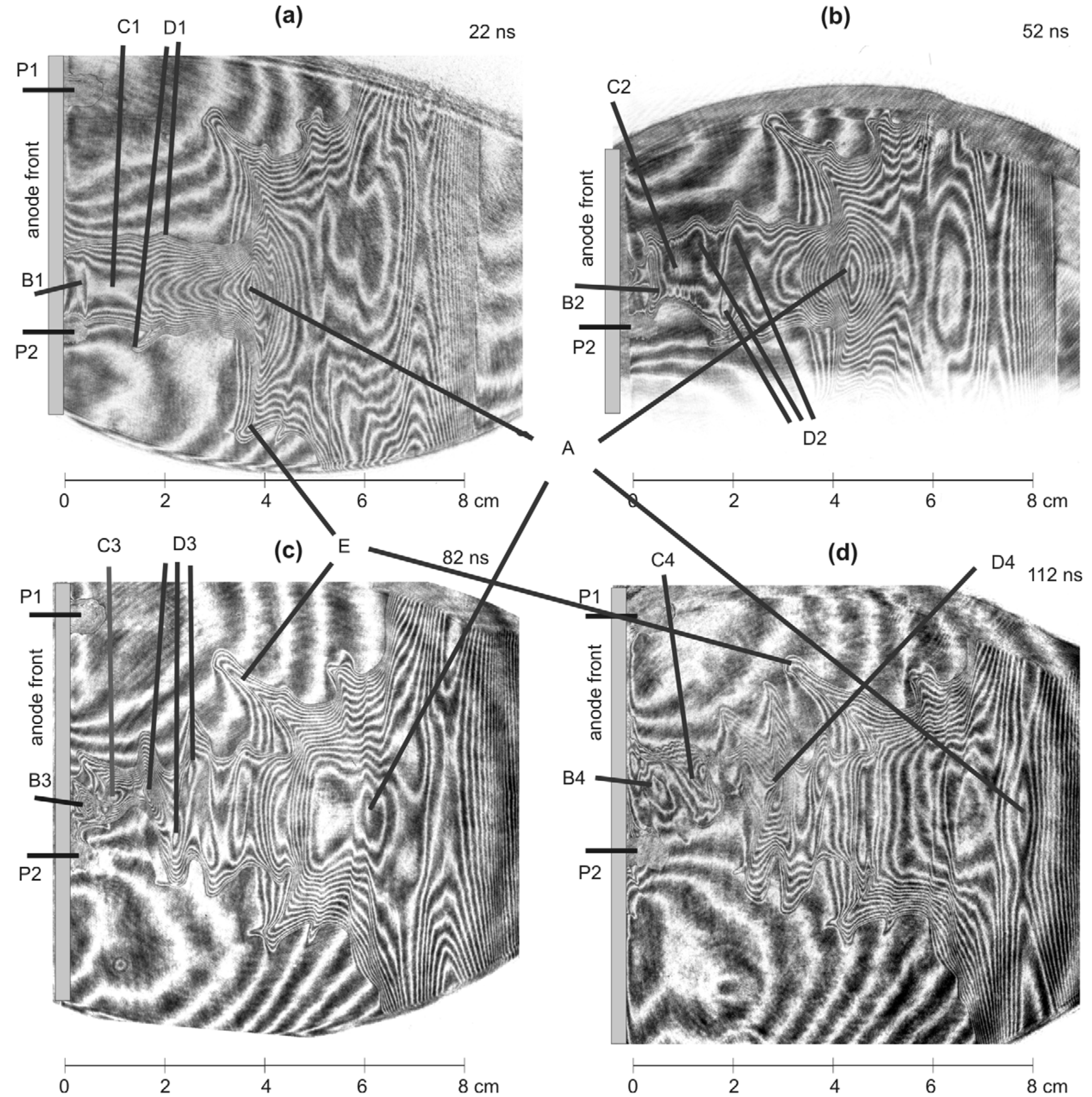

Figure 4. Shot 9367 registered at (*a*) 22 ns, (*b*) 52 ns, (*c*) 82 ns, (*d*) 112 ns: interferograms at the time of instability evolution with transformation of plasmoid A formed during the first neutron pulse, second plasmoid B, toroidal-like coil structure C and helical structure D and dominant lobule E. Here P1 and P2 show the position of the magnetic probes at radial distances of 4 and 1.3 cm.

structures between necks, the plasma density increases and forms toroidal disks. This evolution is similar to that existing during the implosion of the dense plasma layer. In front of the dense localities, the toroidal component of the current, also initially uniform, concentrates into a few toroidal structures, usually in the axial position of the lobules, which existed at the implosion and were reduced during stagnation. B_z increases in the toroidal structures and during implosion of the necks, as the plasma is ejected from the necks to the center of the plasmoids. The maximal plasma density in the plasmoid is reached at the onset of the intense HXR and neutron production.

In four interferograms in figure 4 we can see one example of this evolution (shot 9367). The plasmoid formed during the first neutron pulse is denoted by A. Its motion is imaged in four phases: the motion toward the center of the dominant lobule denoted by E in figure 4(*a*), coalescence with the lobule central range 4 (*b*), penetration through the lobule range 4 (*c*), and penetration through the plasma layer face and formation of the plasma bubble 4 (*d*) usually observed in the interferogram and Schlieren pictures registered in many plasma focus devices. The plasma column transformation is shown at the evolution of the second (B) and third (C) plasmoids, and at the evolution of the helical structure (D). One can see the phases of the second plasmoid evolution: the bird (B1), the formation (B2), the maximum of internal plasma density (B3) and the explosion accompanied by neutron production (B4). We can also observe the transformation of the initially axially uniform part of the column (C1) into a structure with interesting narrow and sharp azimuthal grooves indicating the presence of toroidal currents (C2); next the formation of the third plasmoid (C3), and its

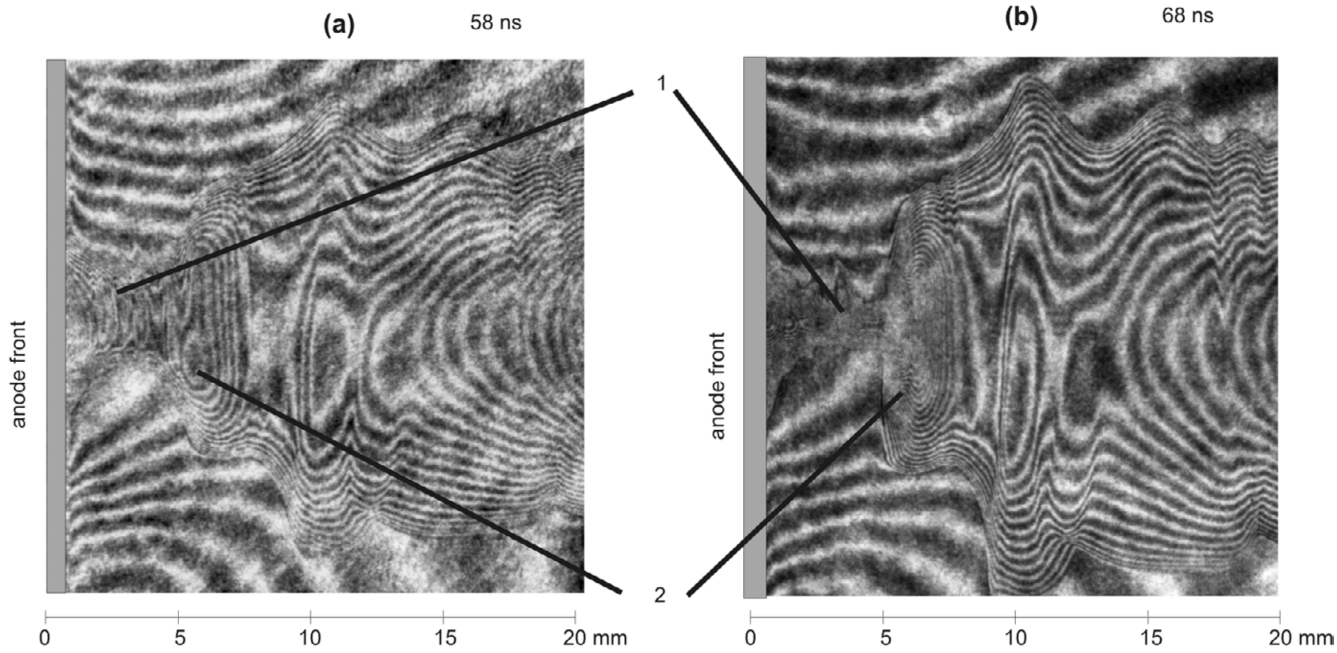

Figure 5. Shot 9119: interferograms showing the evolution of the constriction (1) and the formation of the plasmoid (2); (*a*) the internal structure of the constriction with internal axial periodicity, (*b*) the boundary profile of the constriction and plasmoid with the partially fuzzy fringe pattern.

decay (C4) accompanied by neutron production. The dominant lobule structure E separates two regions of the column, the narrower close to the anode and the wider, close to the umbrella shape of the current sheath.

In figure 5, we show the evolution of the constriction in the interferograms registered in shot 9119. A neck (constriction) is created close to the anode, where the column has a lower plasma density. After partial implosion, the axial disruption of initial homogeneity is imaged (figure 5(*a*)). On the constriction surface the instability $m = 0$ forms on a smaller and finer scale than in the total column. These small and dense plasmoids are formed at the final phase of evolution of each constriction.

In figure 5(*b*) we can see an example of the other typical phenomenon: the partially broken pattern of the fringes in the region of the constriction and plasmoid close to the anode at the time of the transport of the plasma from the neck to the generated plasmoid. In the plasmoid region at a larger distance from the anode, this pattern is clearly apparent. These pictures can illustrate the transformation of the turbulent structure to the stable plasmoid. This phase is accompanied by the production of neutrons with energy close to 2.45 MeV (with energy of mother deuterons of tens of keV).

In figure 6, the densitograms registered in shot 9354 and the scenario of the internal currents in this phase are shown, similarly to the one described in section 3.2. It shows the formation of toroidal current with a stabilized poloidal component. During implosion of the necks, the plasma from the necks is injected inside the central region of toroidal structures, where the plasma density increases. The toroidal structures are transformed first into toroidal disks with maximum density in their axis region and later into plasmoids. Then, the kinetic energy of the plasma jet is transformed through a magnetic dynamo into the magnetic energy of the stronger closed poloidal and toroidal currents.

This configuration of constriction and the plasmoid with the maximum of internal magnetic energy (B_z about 8 T in [11]) is the starting state for their fast decay accompanied by the acceleration of fast electrons and deuterons with energies of hundreds of keV, producing HXRs and neutrons. B_φ on the boundary of the plasma column reaches the value of 15–25 T (estimated for the total current inside the diameter of 1.0–2.0 cm).

3.5. The dominant neutron pulse

During the dominant neutron production, the explosion of the constriction and plasmoid correlates with the decrease in B_z. The changes in the magnetic field during constriction and plasmoid formation, and specifically the fast disintegration correlated with HXR and neutron production, suggest the induction of an electric field contributing to the high-energy charged particle acceleration (figure 7).

4. Discussion and summary

The scenario of the evolution of pinch in a plasma focus discharge was based on the experimental results from the PF-1000 device (at 2 MA in deuterium gas) with neutron, x-ray, interferometry and magnetic probe diagnostics with temporal and spatial resolution. An important role is played by the changes in the axial component of the magnetic field measured during the evolution of toroidal and plasmoidal structures inside the dense plasma column. The axial magnetic field is the component of the poloidal magnetic field connected with the toroidal currents. The formation of these closed currents and the increase in the internal magnetic field correlate with the transport of the plasma into the toroidal and plasmoidal structures from neighboring regions with a

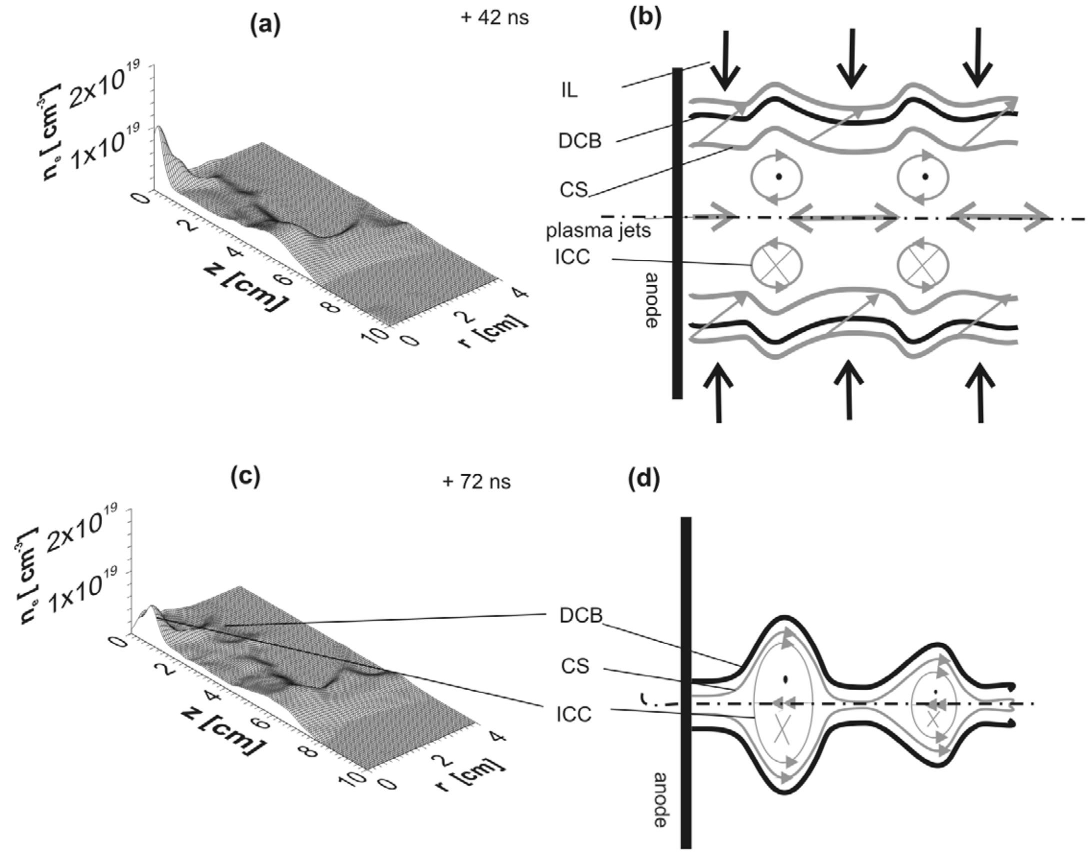

Figure 6. (*a*) Shot 9354: the densitogram of the plasma column at the time of the start of instabilities, (*b*) possible distribution of the current at the start of instabilities, (*c*) shot 9354: densitogram of the plasma column on the threshold of the intense neutron pulse, (*d*) a possible distribution of the current in the plasma column on the threshold of the intense neutron pulse. The imploding layer (IL), the dense column boundary (DCB), the current sheath (CS) and the internal closed current (ICC) are indicated here.

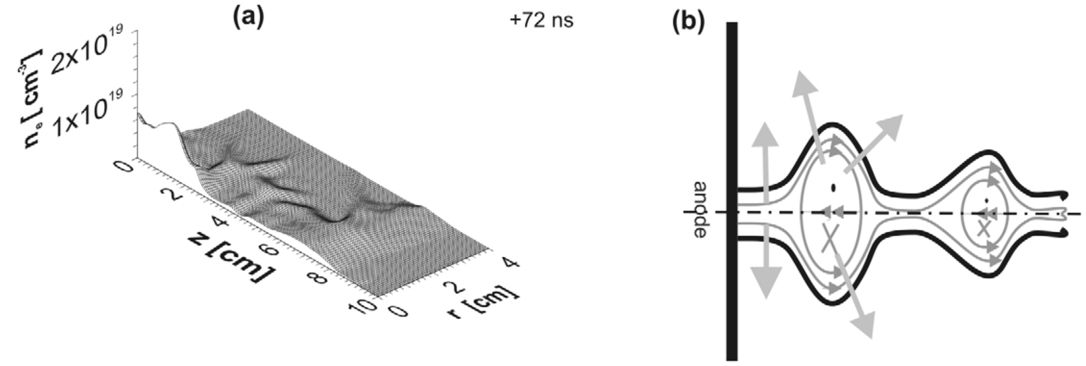

Figure 7. (*a*) Shot 9354: densitogram of the plasma column at the time of dominant neutron production, (*b*) scenario of explosion of the constriction and plasmoid.

higher pinching pressure. These streams of plasma increase the internal magnetic energy in the plasmoids at the expense of kinetic energy, probably through turbulence and the magnetic dynamo effect. This scenario should be completed by the transformation of closed currents and magnetic fields outside the dense pinch, although regarding these phenomena as sufficient experimental evidence is not possible at this time.

As mentioned above, we can quantitatively estimate the magnetic energy in the constrictions and plasmoids that can be released during their decay and we can compare it with the energy of the accelerated deuteron beams. In the plasmoid with a plasma density of $4 \times 10^{24}\,\mathrm{m}^{-3}$, diameter of 2–4 cm, volume of about 1 cm^3 and pinching magnetic field of 15–30 T, we can evaluate the relevant pinching pressure as $(1.7–3.5) \times 10^5$ Pa and the energy absorbed in the plasmoid as $(1.7–3.5) \times 10^2$ J.

The results presented here were obtained for the plasma focus column, but they should be useful here for the *z*-pinch plasma at higher energy densities as well, also due to both deeper analogy of neutron production in two pulses and the similar form of the plasma column in plasma foci and fast *z*-pinches registered with x-frames [15].

Acknowledgments

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